

Numerical Simulation of Financial SDEs

Problem 4 GBM Benchmark and Non-Affine Volatility

Coursework Report

September 2026

Abstract

This report compares exact simulation, Euler–Maruyama (EM), and Milstein for geometric Brownian motion (GBM), and then investigates a non-affine stochastic-volatility model for which a numerical reference is required. All discretisations are coupled through aggregated Brownian increments. For GBM, the measured terminal strong orders are 0.500 for EM and 0.991 for Milstein, consistent with orders $1/2$ and 1 . Weak mean bias is separated from Monte Carlo uncertainty using paired differences, independently fitted control variates, and an analytic discrete-mean identity. The terminal distribution is checked against the exact lognormal law. In the non-affine model, EM is applied to log price and the volatility factor, preserving positive prices. Its observed strong rate is 0.491 on the stated fitting window. Payoff bias is reported with confidence intervals and three nested reference grids; unresolved biases are not assigned a convergence order. The experiments demonstrate why an exact benchmark validates a method, while agreement between numerical grids requires a separate reference-resolution assessment.

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Reproducibility: complete Python source, seeds, numerical tables, figures, and run logs accompany this report. Reported confidence intervals are pointwise Monte Carlo intervals.

1 Introduction and theoretical background

The supplied problem pack selects an SDE pathway in which strong and weak convergence, sampling uncertainty, and nested-grid validation replace deterministic ODE stability-region and adaptive-step analyses [1]. This topic is useful because GBM offers an exact pathwise benchmark, whereas the prescribed non-affine model makes time discretisation part of the solution. The objective is to establish what the experiments actually validate, rather than infer accuracy from plausible-looking sample paths.

1.1 GBM and its exact transition

For $S_0 > 0$, the Itô SDE and log transformation are

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad (1)$$

$$dX_t = (\mu - \frac{1}{2}\sigma^2) dt + \sigma dW_t, \quad X_t = \log S_t. \quad (2)$$

Itô's correction is essential: omitting $-\sigma^2/2$ changes the mean. Integration gives

$$S_t = S_0 \exp\{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t\}. \quad (3)$$

Thus S_T is lognormal, with

$$m = \mathbb{E}[S_T] = S_0 e^{\mu T}, \quad (4)$$

$$v = \text{Var}(S_T) = S_0^2 e^{2\mu T} (e^{\sigma^2 T} - 1), \quad (5)$$

$$q_p = S_0 \exp\{(\mu - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}\Phi^{-1}(p)\}. \quad (6)$$

For the chosen GBM baseline $S_0 = 100$, $\mu = 0.05$, $\sigma = 0.30$, $T = 1$, $m = 105.1271$ and $v = 1040.7868$ (rounded). GBM parameters are experiment choices; the non-affine parameters on page 4 are prescribed by the problem.

1.2 Three methods on a common Brownian path

Let $h = T/N$, $\Delta W_n \sim N(0, h)$. Exact simulation uses

$$S_{n+1} = S_n \exp\{(\mu - \frac{1}{2}\sigma^2)h + \sigma\Delta W_n\}. \quad (7)$$

The price-based EM and scalar Milstein updates are

$$S_{n+1}^E = S_n^E (1 + \mu h + \sigma \Delta W_n), \quad (8)$$

$$S_{n+1}^M = S_n^M \{1 + \mu h + \sigma \Delta W_n + \frac{1}{2}\sigma^2[(\Delta W_n)^2 - h]\}. \quad (9)$$

The extra term follows from $\frac{1}{2}b(S)b'(S)[(\Delta W)^2 - h]$ with $b(S) = \sigma S$ [2]. Under the usual regularity assumptions, terminal strong orders are 1/2 and 1; weak order is 1 for suitable smooth test functions. Applying EM to the GBM log price instead is exact at grid points. Its additive-noise Milstein correction vanishes, so it cannot demonstrate a nontrivial order-one comparison.

2 Monte Carlo methodology

2.1 Coupling and uncertainty

Generate independent paths on the finest grid. A coarse Brownian increment is the sum of the corresponding fine increments, so

$$\Delta W_n^{(h)} = \sum_{j=nr}^{(n+1)r-1} \Delta W_j^{(\delta)}, \quad h = r\delta. \quad (10)$$

For GBM, the exact terminal value uses the same total W_T . For non-affine volatility, coarse and reference states use the same paired Brownian drivers. Independent random numbers across resolutions would introduce a nonvanishing path mismatch and invalidate a strong-order experiment.

For M independent paths, define $A_i(h) = |S_{T,i}^{(h)} - S_{T,i}^{\text{ref}}|$. The terminal L^1 strong-error estimator and its approximate 95% interval are

$$\hat{e}_s(h) = M^{-1} \sum_i A_i(h), \quad \hat{e}_s(h) \pm 1.96 s_A / \sqrt{M}. \quad (11)$$

The observed order is the least-squares slope of $\log \hat{e}_s$ against $\log h$ over an explicitly reported window. Points at different h are correlated because of coupling; the slope is descriptive, and no independent-regression confidence interval is claimed.

For a payoff φ , use signed paired differences

$$D_i(h) = \varphi(S_{T,i}^{(h)}) - \varphi(S_{T,i}^{\text{ref}}), \quad \hat{b}(h) = \bar{D}, \quad \text{CI}_{95} = \bar{D} \pm 1.96 s_D / \sqrt{M}. \quad (12)$$

The weak error is $|\mathbb{E}[D]|$, not $\mathbb{E}|D|$. Signed intervals are retained, including those crossing zero. A log-log trend in an absolute noisy estimate is not sufficient evidence of an order. These intervals quantify sampling error, not the time bias of a numerical reference.

2.2 Resolving GBM weak bias

For both schemes, the conditional expected multiplier is $1 + \mu h$. Consequently

$$\mathbb{E}[S_N^E] = \mathbb{E}[S_N^M] = S_0(1 + \mu h)^{T/h}, \quad (13)$$

$$b_{\text{exact}}(h) = S_0(1 + \mu h)^{T/h} - S_0 e^{\mu T}, \quad (14)$$

$$b_{\text{exact}}(h) = -\frac{1}{2} S_0 e^{\mu T} \mu^2 T h + O(h^2). \quad (15)$$

This identity proves first-order weak bias for the mean in this nondegenerate baseline. It is a validation target, not a Monte Carlo estimate.

The simulation also forms zero-mean control variates C_i and estimates $\overline{D - \beta^\top C}$. The coefficients β are fitted on a separate pilot sample and then held fixed, so the production estimate remains unbiased. Both raw and controlled intervals are retained in the CSV. The pilot is excluded from reported production sample sizes. A short explanation is given on page 6; the five implemented controls and their coefficients are documented in the supplied `gbm.py` and saved results.

3 Model and experimental protocol

3.1 Non-affine stochastic volatility

Write $X = \log S$, and use the prescribed system

$$dX_t = \{\mu - \frac{1}{2}g(Y_t)^2\} dt + g(Y_t) dW_t^{(1)}, \quad (16)$$

$$dY_t = \kappa(\theta - Y_t) dt + \xi \sqrt{1 + Y_t^2} dW_t^{(2)}, \quad (17)$$

$$g(y) = \sigma_{\min} + \frac{\sigma_{\max} - \sigma_{\min}}{1 + e^{-y}}, \quad d\langle W^{(1)}, W^{(2)} \rangle_t = \rho dt. \quad (18)$$

Table 1: Prescribed non-affine benchmark.

S_0	Y_0	μ	κ	θ	ξ
100	0	0.05	2	-0.2	0.6
ρ	$\sigma_{\min}$	$\sigma_{\max}$	T	X_0	strike K
-0.7	0.10	0.50	1	$\log 100$	100

From independent standard normals Z_1, Z_2 , form

$$\Delta W^{(1)} = \sqrt{h}Z_1, \quad \Delta W^{(2)} = \sqrt{h}(\rho Z_1 + \sqrt{1 - \rho^2}Z_2). \quad (19)$$

The implemented EM step is simultaneous at the left endpoint:

$$X_{n+1} = X_n + [\mu - \frac{1}{2}g(Y_n)^2]h + g(Y_n)\Delta W_n^{(1)}, \quad (20)$$

$$Y_{n+1} = Y_n + \kappa(\theta - Y_n)h + \xi \sqrt{1 + Y_n^2} \Delta W_n^{(2)}, \quad (21)$$

$$S_{n+1} = \exp(X_{n+1}). \quad (22)$$

The use of Y_n in both updates matters. Replacing it by Y_{n+1} in the first equation changes the discretisation. The logistic implementation avoids exponential overflow, and finiteness and positive prices are checked during simulation.

3.2 Grids and reproducibility

Table 2: Computational protocol; grids are nested powers of two.

Quantity	Value	Quantity	Value
GBM production paths	160,000	Independent pilot	20,000
GBM seed	2026090804	Pilot seed	2026090805
GBM grids N	8 to 1024	GBM batch	2000
Non-affine paths	100,000	Seed	2026090804
Coarse N	8,16,32,64,128,256	Reference N	2048,4096,8192
GBM elapsed	30.9 s	Non-affine elapsed	107.8 s

All calculations use double precision and fixed NumPy generators. Production paths are processed in batches to bound memory. Seeds, batch sizes, runtime versions, elapsed times, and full precision results are stored in the accompanying JSON files. No external market data are used. GBM compares exact, EM, and Milstein on price; the non-affine experiment uses EM with successively finer references. The payoff is $\varphi(S) = (S - 100)^+$, an *undiscounted expected call payoff* under the stated model drift. No risk-neutral interpretation or option price is inferred without a pricing measure and discount rate.

4 GBM numerical results

4.1 Strong convergence

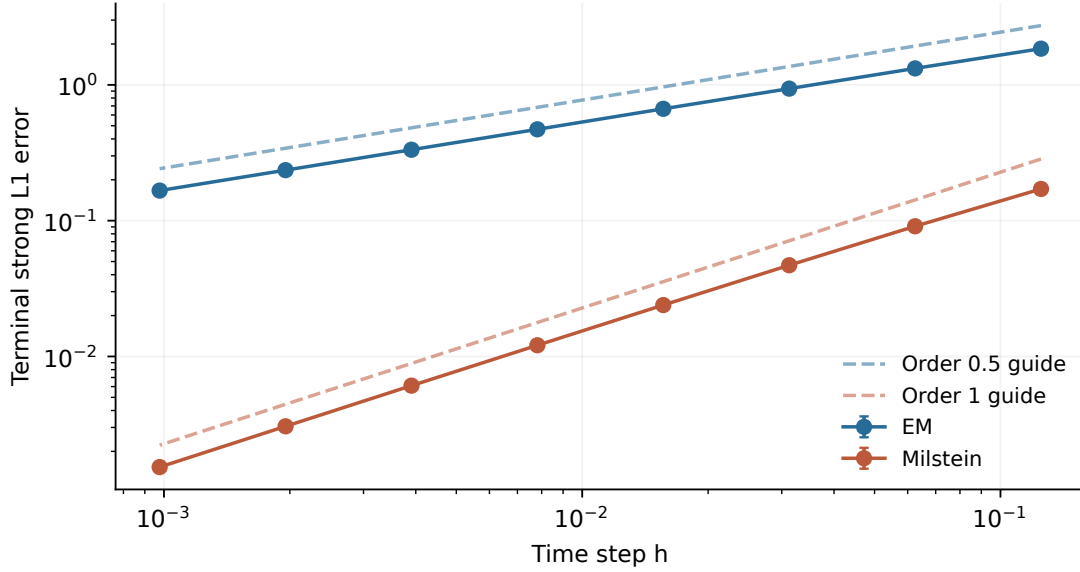


Figure 1: Coupled GBM terminal L^1 error. Error bars are pointwise 95% Monte Carlo intervals; many are smaller than the marker. Reference lines indicate orders $1/2$ and 1 .

Table 3: GBM strong error estimates with 95% interval half-widths.

h	EM	Milstein
1/8	1.84750 ± 0.00809	0.17127 ± 0.00100
1/16	1.32218 ± 0.00562	0.09096 ± 0.00048
1/32	0.93847 ± 0.00396	0.04697 ± 0.00023
1/64	0.66641 ± 0.00278	0.02391 ± 0.00011
1/128	0.47071 ± 0.00196	0.01209 ± 0.00006
1/256	0.33297 ± 0.00139	0.00610 ± 0.00003
1/512	0.23550 ± 0.00098	0.00306 ± 0.00001
1/1024	0.16658 ± 0.00069	0.00153 ± 0.00001

The fitted slopes are 0.500 for EM and 0.991 for Milstein on $N = 64, 128, 256, 512, 1024$. The observed rates match the theoretical distinction: EM omits the quadratic stochastic increment, whereas Milstein accounts for it. At the finest step, Milstein has a substantially lower pathwise error. This comparison uses exactly the same Brownian paths, so differences between methods arise from the update rules rather than from different realizations of the driving noise.

The exact solution has no time-discretisation error at the simulated grid points. Small floating-point changes can occur when sums are regrouped, but these are not a convergence rate. The reported strong errors compare terminal prices rather than the maximum error over the whole path; an experiment using a path-supremum norm would be a different diagnostic.

4.2 Weak convergence and variance reduction

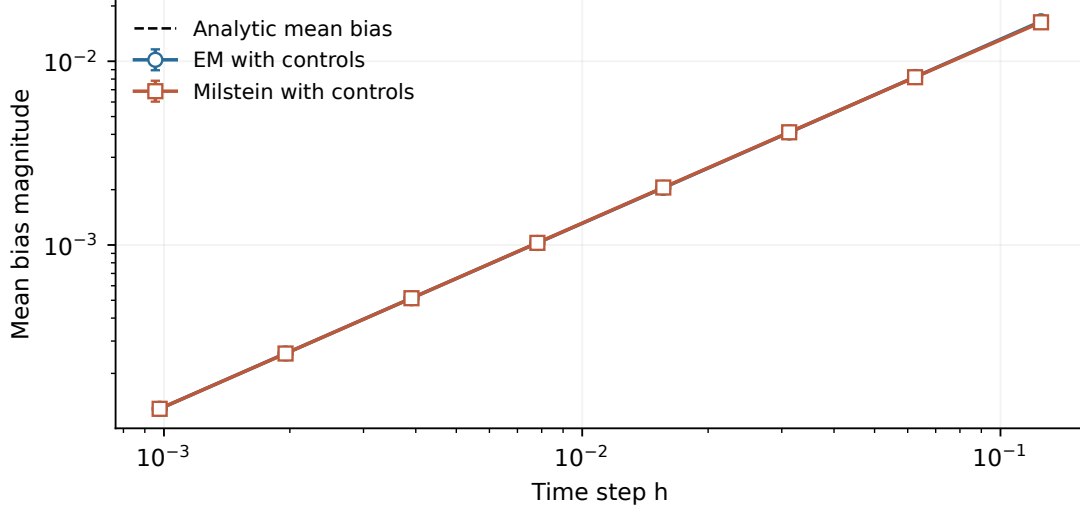


Figure 2: GBM mean bias magnitude. Controlled paired estimates are compared with the analytic discrete-mean bias. Raw estimates and intervals remain available in the numerical tables.

Table 4: Signed weak bias; controlled estimates plus 95% half-widths.

h	Analytic bias	EM controlled	Milstein controlled
1/8	-1.636×10^{-2}	$-1.649 \times 10^{-2} \pm 1.940 \times 10^{-4}$	$-1.629 \times 10^{-2} \pm 9.901 \times 10^{-5}$
1/32	-4.102×10^{-3}	$-4.096 \times 10^{-3} \pm 2.811 \times 10^{-5}$	$-4.106 \times 10^{-3} \pm 1.399 \times 10^{-5}$
1/128	-1.026×10^{-3}	$-1.025 \times 10^{-3} \pm 3.653 \times 10^{-6}$	$-1.027 \times 10^{-3} \pm 1.810 \times 10^{-6}$
1/512	-2.566×10^{-4}	$-2.567 \times 10^{-4} \pm 4.660 \times 10^{-7}$	$-2.566 \times 10^{-4} \pm 2.301 \times 10^{-7}$
1/1024	-1.283×10^{-4}	$-1.284 \times 10^{-4} \pm 1.649 \times 10^{-7}$	$-1.283 \times 10^{-4} \pm 8.144 \times 10^{-8}$

The analytic bias is identical for EM and Milstein even though their strong errors differ. Its fitted order is 0.9993. The controlled Monte Carlo slopes are 0.999 and 1.000 on the stated data, interpreted alongside the intervals rather than as a substitute for equation (14). At small h , the raw paired EM estimate is too noisy to resolve this small bias reliably.

The purpose of a control variate is to remove sampling noise without changing the quantity being estimated. If $D = S_N^h - S_T$ and a random variable C has known mean zero, then, for a fixed coefficient β ,

$$\mathbb{E}[D - \beta C] = \mathbb{E}[D]. \quad (23)$$

When C varies with D , subtracting a suitable multiple can make the sample average much less noisy. The implementation uses five such variables calculated from the same Brownian increments. Their coefficients are fitted on 20,000 separate pilot paths and then held fixed for the 160,000 production paths. The detailed formulas remain in `gbm.py`; they are a computational aid, rather than an additional convergence method.

For EM at $N = 1024$, the raw paired standard error is 5.46×10^{-4} , larger than the analytic bias magnitude 1.28×10^{-4} . After variance reduction it is 8.41×10^{-8} . This explains why variance reduction was used. Milstein's raw paired interval at this grid already excludes zero. The analytic bias is used to check the simulated answer, not substituted for it.

4.3 Distribution and positivity

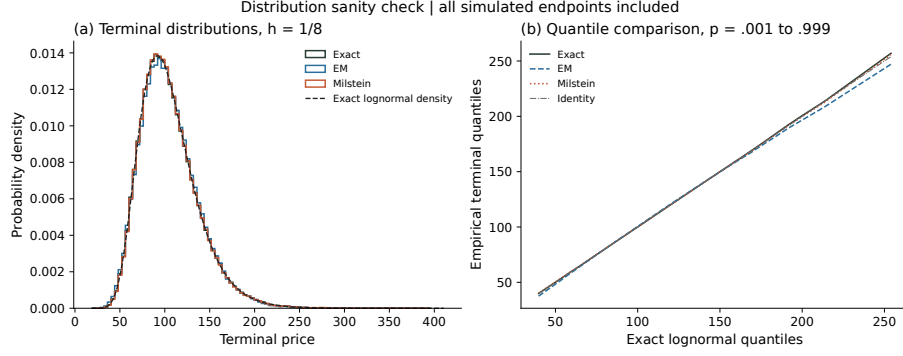


Figure 3: GBM terminal distribution compared with the exact lognormal law. The histogram is a sampling diagnostic; quantiles give a more precise distributional comparison.

Table 5: Terminal quantiles; numerical columns use $h = 1/8$.

p	Lognormal	Exact simulation	EM	Milstein
0.01	50.012	49.928	48.387	50.213
0.05	61.357	61.378	60.512	61.615
0.25	82.091	82.151	82.303	82.263
0.50	100.501	100.557	101.088	100.582
0.75	123.041	123.164	123.653	123.057
0.95	164.618	164.485	163.370	164.043
0.99	201.961	202.402	198.376	201.539

The exact-simulation mean is 105.1987 and its variance is 1045.0120; the mean interval is [105.0403, 105.3571]. The analytic mean is 105.1271 and variance is 1040.7868. Sampling fluctuations in empirical moments and quantiles remain even for exact simulation; they must not be described as discretisation error.

For $S_n > 0$, EM becomes negative with one-step probability

$$\mathbb{P}(S_{n+1}^E < 0 \mid S_n > 0) = \Phi\left(-\frac{1 + \mu h}{\sigma\sqrt{h}}\right). \quad (24)$$

For Milstein, completing the square gives the multiplier

$$\frac{1}{2}\sigma^2(\Delta W + 1/\sigma)^2 + \frac{1}{2} + (\mu - \frac{1}{2}\sigma^2)h. \quad (25)$$

It can be negative only if $(\sigma^2 - 2\mu)h > 1$. Here $\sigma^2 - 2\mu = -0.01$, so Milstein is positive for every $h > 0$ in this parameter set. EM has a nonzero theoretical probability of a negative update, although that probability is extremely small for these parameters.

Table 6: Observed GBM negative terminal frequency: 160,000 paths for each method at each grid.

N	EM: $S_N < 0$	Milstein: $S_N < 0$
8, 16, 32, 64	0/160,000 (0%) at each grid	0/160,000 (0%) at each grid
128, 256, 512, 1024	0/160,000 (0%) at each grid	0/160,000 (0%) at each grid

The saved counts are for $S_N \leq 0$ and are all zero, so the strictly negative counts in Table 6 are also zero. These are terminal frequencies, not counts of paths ever becoming negative: intermediate GBM prices were not recorded by this diagnostic. Zero observed negatives does not prove that EM preserves positivity. Negative prices could distort payoff estimates; clipping them to zero would also change the method. Neither scheme is clipped here.

5 Non-affine numerical results

5.1 Strong differences against a numerical reference

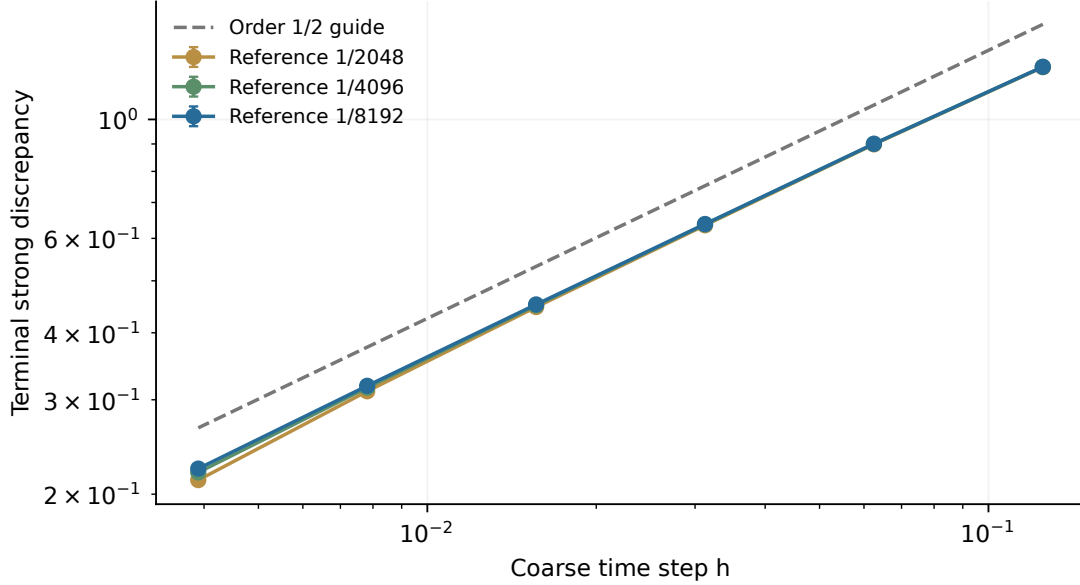


Figure 4: Terminal strong differences against three coupled EM reference grids. The reference is numerical, so these are not exact-error curves.

Table 7: Strong discrepancy against $h_{\text{ref}} = 1/8192$; 95% half-widths.

h	$\mathbb{E} S_h - S_{\text{ref}} $	Last reference difference / error
1/8	1.25354 ± 0.00702	3.2%
1/16	0.90087 ± 0.00489	4.5%
1/32	0.63818 ± 0.00341	6.3%
1/64	0.45176 ± 0.00241	8.9%
1/128	0.31854 ± 0.00170	12.6%
1/256	0.22323 ± 0.00119	18.0%

Against the finest reference, the empirical slope is 0.491 over $N = 8, 16, 32, 64$. The result is consistent with approximately half-order strong behavior on this window. Curves bend when a coarse grid approaches a reference grid: two nearly identical discretisations can agree well while both retain a nonzero error relative to the continuous SDE.

If S^* denotes the unknown exact terminal value, the reverse triangle inequality yields

$$|\mathbb{E}|S_h - S^*| - \mathbb{E}|S_h - S_r|| \leq \mathbb{E}|S_r - S^*|. \quad (26)$$

The right-hand side is unknown. Replacing r by a finer grid measures sensitivity of the benchmark, not that unknown error itself. The next sections therefore report reference changes separately and avoid presenting a rate fitted through the finest coarse points as exact verification.

5.2 Weak payoff differences

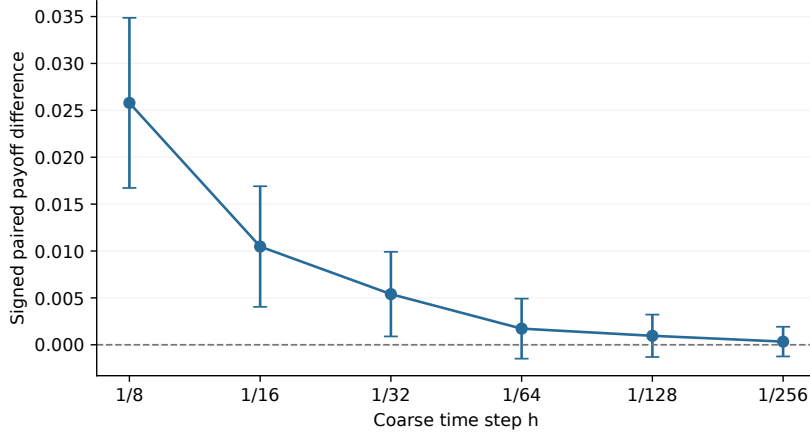


Figure 5: Signed paired differences in the expected call payoff relative to the finest EM reference. Error bars are 95% sampling intervals; the zero line identifies unresolved bias signs.

Table 8: Signed paired payoff differences against $h_{\text{ref}} = 1/8192$.

h	Estimated bias	95% interval	Zero excluded
1/8	0.02579	[0.01672, 0.03486]	Yes
1/16	0.01048	[0.00405, 0.01691]	Yes
1/32	0.00540	[0.00089, 0.00991]	Yes
1/64	0.00172	[-0.00148, 0.00492]	No
1/128	0.00096	[-0.00130, 0.00322]	No
1/256	0.00034	[-0.00124, 0.00192]	No

The 95% intervals include zero at $N = 64, 128, 256$. No weak order is assigned to the complete grid range. The finest-reference payoff is estimated as 14.53804 with interval [14.40515, 14.67093]. The payoff has a corner at the strike, so a weak order should not simply be assumed from smooth-function theory. The paired intervals show whether the estimated difference can be distinguished from sampling noise; they do not include reference bias.

5.3 Why use a call payoff instead of just the mean?

The continuous model satisfies $\mathbb{E}[S_T] = 100e^{0.05}$. However, the log-EM method also gives this mean at *every* step size. To see why, set $g_n = g(Y_n)$. At the start of the step, S_n and g_n are known and the next increment is $N(0, h)$. Hence

$$\begin{aligned} \mathbb{E}[S_{n+1} \mid \mathcal{F}_{t_n}] &= S_n e^{(\mu - g_n^2/2)h} \mathbb{E}[e^{g_n \Delta W_n^{(1)}} \mid \mathcal{F}_{t_n}] \\ &= S_n e^{(\mu - g_n^2/2)h} e^{g_n^2 h/2} = S_n e^{\mu h}. \end{aligned} \tag{27}$$

Here $\mathcal{F}_{t_n}$ means the information available at the start of the step. Repeating this calculation gives $\mathbb{E}[S_N] = 100e^{0.05}$. Thus the mean is a useful implementation check, but has zero time-stepping bias for this scheme and cannot reveal its weak convergence rate.

Equal means do not imply equal expected payoffs. For example, a price always equal to 100 and a price equally likely to be 80 or 120 both have mean 100. Their expected call payoffs $(S - 100)^+$ are respectively 0 and 10. The call payoff therefore tests more of the terminal distribution than its mean. Unlike the mean, its expectation is not automatically preserved by the log-EM update, so changing the step size can produce a genuine weak bias. This makes it a nontrivial weak observable; it is not the only possible choice, and does not guarantee a measurable bias at every grid.

6 Reference refinement and validation

Table 9: Coupled refinement of the numerical reference.

Refinement $N_a \rightarrow N_b$	Strong difference	Signed payoff difference and 95% CI
2048 $\rightarrow$ 4096	0.05679 ± 0.00030	-0.000023 $[-0.000423, 0.000377]$
4096 $\rightarrow$ 8192	0.04016 ± 0.00021	0.000094 $[-0.000188, 0.000376]$

The average absolute difference falls from 0.05679 for 2048 versus 4096 steps to 0.04016 for 4096 versus 8192 steps. To judge its scale, divide 0.04016 by the coarse-to-finest strong difference: the ratios are 3.2%, 4.5%, 6.3%, and 8.9% for $N = 8, 16, 32, 64$, increasing to 12.6% and 18.0% for $N = 128, 256$. Reference refinement therefore matters more, in relative terms, at the finer coarse grids. We retain the original $N = 8, 16, 32, 64$ fitting window and report its slope as empirical. The earlier report used 10% as an author-chosen rule of thumb; it is not a requirement in the problem pack or an established tolerance justified by the cited references. We do not use it as a pass/fail test here. The observed changes provide refinement evidence, not a bound on error relative to the unknown exact solution.

Table 10: Brownian and state diagnostics at the finest non-affine grid.

Quantity	Observed	Target or scope
Increment variance 1	1.221×10^{-4}	1.221×10^{-4}
Increment variance 2	1.221×10^{-4}	1.221×10^{-4}
Increment correlation	-0.699675	-0.700000
Sampled increments	4,194,304	Predetermined first batch
Total path updates	1,484,000,000	All grids
Finite intermediate states	Pass	X and Y
Positive finite prices	Pass	All observed states

The variance targets for both Brownian increments are h , and their target correlation is -0.7 . The diagnostics are calculated from the generated fine increments, not inferred from the generator formula. State checks include intermediate steps, because a finite terminal value alone would not rule out an invalid intermediate update.

The finest-grid mean is 105.08796 with 95% interval $[104.89962, 105.27629]$. The corresponding undiscounted payoff estimate is 14.53804 with interval $[14.40515, 14.67093]$. This latter interval describes sampling uncertainty about the *finest discrete model*; the refinement table is still needed to assess its bias.

As requested in the problem pack, EM is used for this two-noise model. A multidimensional Milstein method would need additional cross-noise terms; the scalar GBM formula cannot simply be copied into this system. No such extension is needed for the present comparison.

7 Discussion and conclusion

The GBM experiments establish three distinct facts. First, coupling against its exact solution recovers the expected strong-order advantage of Milstein. Second, strong order and weak order answer different questions: both methods have the same first-order bias in the terminal mean despite different pathwise errors. Third, distributional and positivity checks add information that a single mean does not contain. Log-price simulation is preferable when exact GBM transitions are available, while price-based discretisation remains a useful controlled validation exercise.

For the non-affine model, bounded positive volatility and the exponential price representation give finite, positive simulated prices on the tested paths. Nested-grid comparisons provide evidence of refinement convergence and expose how a numerical benchmark can affect the measured rate. They do not produce exact errors. In particular, payoff intervals that overlap zero cannot support an asserted weak slope, and a reference residual comparable with a coarse-grid difference limits the precision of that comparison.

The main computational trade-off is between path count and temporal resolution. Halving h approximately doubles updates per path. Increasing M by a factor of four approximately halves the Monte Carlo standard error. Coupled differences and control variates address the sampling component, while refinement addresses discretisation. Extra paths cannot repair an underresolved reference, and smaller steps cannot by themselves remove sampling uncertainty.

The main experiments follow the SDE-specific route of Problem 4: exact–EM–Milstein comparison, strong and weak convergence with coupled paths, lognormal validation, sampling intervals, and non-affine nested-reference checks. No deterministic stability plot or scalar Milstein extension is substituted for these requirements. The optional positivity diagnostic reports terminal frequencies; it does not count every negative intermediate update. Conclusions apply to the stated parameters and grid ranges.

Reproducibility

The Python files and saved results in the original package document the random seeds, sample sizes and updates used. This revised report reuses those experiments; it does not present them as a new simulation run. The appendix records the core updates and reproduction commands.

AI acknowledgement

OpenAI Codex was used to help explain the mathematical methods, develop and check the Python code, carry out numerical calculations, and draft and revise the report. This assistance included improving the clarity of the discussion and presentation of the results.

References

- [1] *Problem Pack: Five Suggested Directions for the ODE IVP Project*. Supplied coursework document, Direction 4, pp. 6–7.
- [2] D. J. Higham. An algorithmic introduction to numerical simulation of stochastic differential equations. *SIAM Review*, 43(3):525–546, 2001. doi:10.1137/S0036144500378302.
- [3] P. E. Kloeden and E. Platen. *Numerical Solution of Stochastic Differential Equations*. Springer, 1992. doi:10.1007/978-3-662-12616-5.

A Python implementation and reproduction

The following listing states the core mathematical operations. The executable modules additionally implement batching, control-variate pilot fitting, uncertainty estimates, diagnostics, exports, and plots; their complete source is supplied alongside this report.

Listing 1: Core SDE updates and coupled error statistics.

```
1 import numpy as np
2
3 def coarsen(dw_fine, n_coarse):
4     m, n_fine = dw_fine.shape
5     assert n_fine % n_coarse == 0
6     return dw_fine.reshape(m, n_coarse, -1).sum(axis=2)
7
8 def gbm_terminal(dw, s0=100., mu=.05, sigma=.3, T=1.):
9     h = T / dw.shape[1]
10    exact = s0 * np.exp((mu-.5*sigma**2)*T
11                      + sigma*dw.sum(axis=1))
12    em = s0 * np.prod(1 + mu*h + sigma*dw, axis=1)
13    mil = s0 * np.prod(1 + mu*h + sigma*dw
14                      + .5*sigma**2*(dw*dw-h), axis=1)
15    return exact, em, mil
16
17 def log_em_step(x, y, dw1, dw2, h):
18     # Values on both right-hand sides are at the old time.
19     sigmoid = np.exp(-np.logaddexp(0., -y))
20     g = .1 + .4*sigmoid
21     x_new = x + (.05-.5*g*g)*h + g*dw1
22     y_new = y + 2*(-.2-y)*h + .6*np.hypot(1.,y)*dw2
23     return x_new, y_new
24
25 def mean_ci(values):
26     m = len(values)
27     mean = np.mean(values)
28     half_width = 1.96*np.std(values, ddof=1)/np.sqrt(m)
29     return mean, mean-half_width, mean+half_width
30
31 # Brownian drivers from independent standard normals:
32 # dw1 = sqrt(h)*z1
33 # dw2 = sqrt(h)*(rho*z1 + sqrt(1-rho**2)*z2)
34 # Strong: mean_ci(abs(s_coarse-s_reference))
35 # Weak: mean_ci(maximum(s_coarse-100,0)
36 #               -maximum(s_reference-100,0))
```

Run the experiments from the original package directory using Python 3.12:

```
python -m pip install -r requirements.txt
python run_all.py

# To validate supplied results without resimulation:
python verify.py
```

Compile this revised report separately, from `undergraduate_revision/`:

```
tectonic Coursework_Undergraduate.tex
```

In the original package, `results/` contains machine-readable tables and logs; `figures/` contains vector PDFs and PNGs. The supplied PDF is already compiled. The appendix listing is explanatory; the tests exercise the actual imported implementation functions. Full reproduction includes both simulation modules, while rebuilding only the report tables does not resimulate paths.